EC107: Microeconomics

In-class Exercises 2 — The Market Forces of Supply & Demand

Covers Day 3. For each **concept check**: vote, discuss with a neighbour, vote again. Short exercises are timed (5 minutes each).

A. Concept checks

C1. The price of tea rises. As a *direct* result, in the market for tea we observe:
(a) a leftward shift of demand (b) a movement *along* demand
(c) a rightward shift of demand (d) a rightward shift of supply

C2. Tea and sugar are complements. The price of tea falls sharply. In the market for *sugar*:
(a) demand shifts left (b) demand shifts right
(c) supply shifts right (d) a movement along demand

C3. Generous subsidies raise the demand for electric cars; at the same time, cheaper batteries raise supply. The equilibrium *quantity* of electric cars:
(a) definitely rises (b) definitely falls
(c) stays the same (d) is ambiguous without magnitudes

C4. At the current price, quantity supplied exceeds quantity demanded. We expect price to:
(a) rise (b) fall (c) stay put (d) be indeterminate

B. Short exercises

N1. (Equilibrium & a shift) Demand: $P = 20 - 2Q$. Supply: $P = 2 + Q$.

(i) Find the equilibrium price and quantity.

(ii) A rise in incomes shifts demand to $P = 26 - 2Q$. Find the new equilibrium.

(iii) State the direction of change in P^* and Q^*.

N2. (Market demand) Two consumers. Maya: $Q_d = 10 - P$. Noah: $Q_d = 16 - 2P$.

(i) For prices low enough that both buy, write market demand as Q in terms of P.

(ii) At what price does Noah drop out of the market?

(iii) Find total quantity demanded at $P = £3$.

C. Discussion

D1. "Whenever there is a shortage of a good, it must mean the world is running out of it." Evaluate this claim using the distinction between *scarcity* and a *shortage*. Give one real example of a shortage that disappeared *without* any increase in the physical stock of the good.